EDUCATION POLICY & MACRO-FISCAL RECONCILIATION DOCUMENT

Kosovo Education Research

The Enthusiasm–Achievement Paradox: An Economic & Policy Study

An inquiry into why Kosovo's Grade 4 students display some of the highest learning motivation globally, yet wrestle with a structural infrastructure that chokes cognitive skill conversion. A multi-pillar analysis of post-conflict legacy, multi-shift schooling bottlenecks, and cosmetic policy mimicry.

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DATA BASELINE

IEA PIRLS 2021 & TIMSS 2023

The Tri-Discipline Paradox

Quantifying the structural mismatch between student engagement and cognitive mastery across Reading, Math, and Science.

Literacy (PIRLS 2021)

85%

Students Who Like Reading

421

Average Achievement Score

High emotional engagement vs. systemic failure to convert interest into reading mastery. Only 27% reach the intermediate benchmark.

Mathematics (TIMSS 2023)

74%

Students Who Like Math

451

Average Achievement Score

The system's highest score, yet a critical underlying confidence gap exists: only 31% of students express high math confidence.

Science (TIMSS 2023)

70%

Students Who Like Science

403

Average Achievement Score

The lowest baseline performance of all three cohorts, marked by a severe 8-percentage-point decline in intermediate benchmarks.

5th-to-95th Percentile Delta

278

Average Achievement Score

SD - 84

Systemic Bifurcation

The stark delta separating the 5th percentile (271) from the 95th percentile (549) exposes a deeply polarized educational baseline. Driven by an exceptionally high standard deviation of 84 points, primary school teachers are forced to navigate an extreme, unmanageable spectrum of student learning needs within the exact same classroom. This structural fragmentation means teachers cannot deliver targeted instruction, stranding a struggling majority while failing to cultivate an advanced elite.

PIRLS Reading Benchmark Distribution

- **0%** Advanced (≥ 625)
- **5%** High (≥ 550)
- **27%** Intermediate (≥ 475)
- **68%** Low to Below Low (< 475)

Critical Threat: **38% of Kosovo's Grade 4 students score "Below Low" (≤ 400),** rendering them functionally illiterate by international standards. The system displays a massive floor restriction with 6% of science and 4% of math cohorts scoring too low for reliable estimation.

Sources: IEA PIRLS Portal (pirls2021.org) | IEA TIMSS Portal (timss2023.org) | World Bank Human Capital Index (humancapital.worldbank.org) | Oral History Kosovo Archives (oralhistorykosovo.org).

The Higher-Order Thinking Deficit

When mapped against Bloom's Taxonomy, Kosovo's primary education system operates effectively at the baseline levels of remembering and understanding, but faces a steep drop-off at applying, analyzing, and evaluating.

Students exhibit a distinct **locate but cannot think** profile across reading, mathematics, and science. They can reliably identify explicit information sitting on a page, but their cognitive capacity collapses the moment an international assessment requires them to independently interpret, integrate, or manipulate that data.

Quantitative Evidence of Cognitive Drop-Offs

The empirical footprint of this deficit is uniform across all three tested disciplines. In PIRLS reading, students score **424** (**+4** relative to their baseline) on straightforward retrieval, but plummet to **412 (−9)** on text interpretation and evaluation. In TIMSS mathematics, students demonstrate a relative grasp of routine numbers and shapes **(+6)**, but experience a massive collapse to **429 (−22)** in data and measurement, where abstract analysis is mandatory. Similarly, in science, the most conceptually demanding domain, Life Science, registers the system's worst performance at **387 (−16)**, completely undercutting relative strengths in basic physical and earth science facts.

Structural Disconnect in Classroom Materials

This deficit is further institutionalized by a severe decoupling between high-level state policies and daily learning materials. While Kosovo's national curriculum mandates competency-based learning, the state-approved textbooks placed on desks remain deeply traditional and flat. These materials are structured almost exclusively around factual definitions, choral repetition, and straightforward recall. Because textbooks do not provide the interconnected cognitive environments or data charts required to practice evaluation, teachers default to a lecture-and-transcribe format, leaving students completely unexposed to analytical tasks.

The Mechanics of Word-Level Overload

The primary engine of this interpretive failure is a working memory bottleneck. Before a child can evaluate a text or problem, basic decoding and reading fluency must be automatic. Because a significant portion of Kosovo's Grade 4 cohort has low automaticity, sub-fluent readers exhaust their entire cognitive bandwidth simply parsing sentences or reading words. With their working memory completely occupied by word-level processing, students have no remaining cognitive space for higher-order meaning-making, schema synthesis, or contextual evaluation.

The Assessment Trap and Background Deficits

Finally, the classroom environment reinforces a false sense of security. National examinations emphasize factual recall, causing teachers to rationally optimize their instruction for rote memorization. This creates a massive gap between local competence (where **87%** of students believe lessons are clear) and international competence tested by PIRLS and TIMSS. This structural trap is compounded by an underlying background knowledge deficit; since interpretation requires prior exposure to varied concepts, the country's high-poverty profile restricts the vocabulary and world knowledge children need to bridge text to deeper meaning.

Sources: IEA PIRLS Portal (pirls2021.org) | IEA TIMSS Portal (timss2023.org) | World Bank Human Capital Index (humancapital.worldbank.org) | Oral History Kosovo Archives (oralhistorykosovo.org).

The Socioeconomic Envelope

Socioeconomic status (SES) operates as the single most powerful predictor of student outcomes in Kosovo, forming a rigid constraint long before children enter a classroom.

Kosovo's macroeconomic landscape not only limits the institutional purchasing power but also dictates the compressed physical duration of the school day, heavily penalizes low-income households, and weakens the structural incentives required to turn motivation into advanced cognitive skills.

The Dominant SES Achievement Gap

The empirical data shows that household socioeconomic background eclipses gender, school infrastructure variation, and pandemic learning losses combined. Across international assessments, only **16%** of students belong to high-SES households, while a massive **33%** to **36%** sit in the lowest socioeconomic bracket. This divide manifests as a devastating **80-point** achievement gap in PIRLS reading and an **85-point** gap in TIMSS mathematics. High-SES students score an average of 510 in math, reaching international baseline competency, whereas low-SES peers stall at 425, trapped below the intermediate threshold due to a lack of language-rich home environments and academic support frameworks.

The Institutional Wage-Allocation Trap

The universal scarcity of school resources documented across 95% of the student population is a direct product of an inflexible national budgeting architecture. While Kosovo commits a respectable 4.6% of its GDP to public education, up to **90.6%** of this recurrent funding is entirely consumed by administrative and teacher wage compensation. This extreme wage absorption leaves the state budget starved of non-wage capital inputs. Consequently, there is virtually zero fiscal space for preventative facility maintenance, baseline digital learning technologies, textbook upgrades, or modern science labs, forcing schools to run on a stripped baseline where resource variations fail to predict student scores.

The Multi-Shift Hour Compression

Because capital funding is choked by the wage-compensation bill, school infrastructure expansions have completely stalled against urban demographic shifts. To cope, the vast majority of public schools operate on a rigid two-shift and occasionally three-shift rotation system. This physical asset deficit forces lessons to be compressed into rushed 40-minute blocks to cycle cohorts through the building. These truncated hours systematically evacuate higher-order learning opportunities. Teachers lack the time required for analytical discussions, experiments, or active retrieval exercises, defaulting to rapid rote dictation as the most time efficient mechanism to deliver data before the bell rings.

The 5.3-Year "Phantom Schooling" Leak

According to the World Bank's Human Capital Index, a child born in Kosovo will achieve only 57% of their potential economic productivity due to systemic educational inefficiencies. While families highly prize the social status of schooling, driving an expected 13.2 years of total seat-time and an 87% sense of school belonging, the actual cognitive mastery tells a profoundly different story. When adjusted for real learning, children lose **5.3 years** of functional schooling, graduating with an effective output of just 7.9 years. The state pays for over a decade of physical attendance, but receives less than eight years of cognitive skill return, illustrating a massive motivation trap where children value the credential but the system simulates the learning.

Sources: IEA PIRLS Portal (pirls2021.org) | IEA TIMSS Portal (timss2023.org) | World Bank Human Capital Index (humancapital.worldbank.org) | Oral History Kosovo Archives (oralhistorykosovo.org).

The COVID Disruption & Fragile Baselines

The COVID-19 pandemic exposed the critical vulnerability of an educational infrastructure heavily reliant on physical classroom presence. While total school closures were structurally shorter than many European peers, the rolling, unpredictable nature of the lockdowns shifted a fragile system into completely passive remote mediums, accelerating the rote learning trap.

38%

Operational Pacing Loss

Of public schools lost 2 to 4 weeks in unpredictable micro-bursts, shattering the possibility of cohesive reading and math tracking.

91% Total Schools

Universally disrupted nationwide by constant, erratic structural shifts between Scenario A, B, and C frameworks.

40%

Device Deprivation

Of Grade 4 students had zero functional device access for formal schoolwork, relying heavily on parents' personal smartphones.

388 Score

Digital Disconnect

Average score of heavy device users, showing that unguided technology access translated to screen-time rather than active learning.

16%

Catastrophic Loss

Of students were "adversely affected a lot," completely falling off the academic grid with unrecovered, permanent baseline skill fade.

-16 Points

Life Science Deficit

The massive deficit driven by the complete abandonment of physical demonstrations and hands-on experiments.

The Chaos of Rolling Continuity Scenarios

Following the March 2020 lockdown, MESTI toggled between Scenario A (in-person), Scenario B (hybrid), and Scenario C (remote). This constantly fractured structural pacing, forcing 38% of public schools to lose 2 to 4 weeks in unpredictable micro-bursts. This system-wide instability disrupted a massive 91% of classrooms nationwide, preventing the tracking of foundational skills.

The Digital Access and Device Disparity

While general internet penetration exceeded 90%, real educational utility was blocked by severe device scarcity. A massive 40% of fourth graders never used a dedicated device, borrowing parents' phones. This explains why heavy device users scored the lowest, averaging 388 points, as devices served as unstructured entertainment rather than guided tools.

One-Way Broadcasting and the Passive Medium

Televised lectures via RTK and YouTube replicated the passive home-schooling of the 1990s. A teacher read information to a camera with zero interaction, formative assessments, or active retrieval. This structure simulated an illusion of learning. Student interest remained superficially high, while cognitive retention collapsed.

Catastrophic Learning Loss on the Floor

Without digital tracking or immediate remediation, the pandemic wiped out fragile phonics gains. The 16% "affected a lot" fell off the grid. Science suffered severely; remote learning forced teachers to abandon physical experiments, driving an 8% drop in intermediate benchmarks and a -16 deficit in Life Science.

Sources: IEA PIRLS 2021 COVID Impact Report (0-3_P21-school-impacted-covid.xlsx) | IEA TIMSS 2023 Achievement Trends Hub (2-1-12_ach-g4s-bmk-trend.xlsx) | UNICEF Learning Passport Evaluation Brief (shkollat.org documentation).

The Policy Landscape & Mechanisms of Cosmetic Reform

The central crisis of Kosovo's educational governance does not stem from a lack of progressive legislation, rather, it is driven by a deep institutional disconnect where sophisticated, European-aligned policies exist on paper but completely fail to change the instructional core inside classrooms. This pattern of cosmetic compliance has systematically institutionalized the rote learning trap.

Isomorphic Mimicry & Western Trap

Following independence, geopolitical pressure drove Kosovo to adopt **isomorphic mimicry** - the practice of copying successful foreign systems to claim legitimacy without the capability to execute. MESTI implemented rapid structural leaps, replacing content-driven frameworks with an advanced Competency-Based Curriculum in **2011** and **2016**. This prioritized analytical outcomes on paper, but was introduced into a system where teachers and resources were entirely unprepared, rendering reforms purely cosmetic.

Two-Track Curriculum Failure

This cosmetic alignment created a severe structural decoupling. While the Core Curriculum demanded critical evaluation, the state-approved subject syllabi in **2018** and traditional textbooks focused heavily on factual definitions and rote memorization drills. Overwhelmed teachers default to treating flat, recall-heavy textbooks as a script to memorize. This failure matches the cognitive paradox where students score high on straightforward retrieval (**+4**) but plummet in interpretive evaluation (**−9**).

Development Vacuum & False Accountability

Teacher preparation and licensing reinforce traditional lecture pedagogy. Pre-service university programs are rigid, and in-service training operates as a credit-accumulation game of short weekend seminars. There is virtually zero classroom-level monitoring to verify modern pedagogy. This lack of supervision allows poor habits to persist while maintaining a false sense of success, as teachers report **69%** satisfaction and students report **87%** clarity. Simple lectures feel clear but fail to develop critical thinking, trapping reading at a flat **421** baseline.

Pre-Primary Mandate & Deficits

To narrow socioeconomic gaps, the national Education Strategy **2022–2026** made pre-primary education (Grade 0) compulsory. However, implementation has collided with a severe infrastructure deficit. Public preschools suffer from extreme space shortages, forcing primary schools to absorb young cohorts into existing facilities. This has worsened overcrowding and accelerated multi-shift rotation. Lacking early childhood tools, play-based environments, or aides, teachers default to desk-bound rows, pushing the rote learning trap a year earlier.

Donor Dependency & Tech Fragmentation

Because **90.6%** of Kosovo's recurrent education budget is locked entirely into administrative wages, the state is structurally dependent on international donors to fund system improvements. The national strategy relies on external loans and grants totaling over **320 million euros** from partners like the World Bank, UNICEF, the EU, and USAID. This funding mechanism splits the policy landscape into fragmented, parallel pilot projects that rarely scale to transform the core national curriculum.

A clear example is the rollout of the e-learning platform shkollat.org via donor-supported emergency initiatives. While technically sophisticated, the platform was launched into a vacuum of teacher digital readiness. Teachers were trained to log in but not how to use technology as an interactive cognitive tool. This explains why unsupervised digital device users see performance collapse to a dismal **388** average score; without active digital pedagogy, technology access simply defaults to passive screen-time distraction.

Sources: Kosovo National Education Strategy 2022–2026 (MESTI) | World Bank Kosovo Public Expenditure Review | IEA PIRLS 2021 & TIMSS 2023 National Contextual Data (5-2-1-3_con-stu-icm.xlsx, 4-2-11-15_con-tea-sos.xlsx, 7-5_students-use-digital-devices.xlsx).

Brain Drain, Remittances, and the Incentive Deficit

To understand why student motivation fails to convert into cognitive proficiency, we must analyze the macro-labor conditions shaping the horizons of Kosovo's youth. The structural crisis of the political economy is defined by a deep decoupling of educational merit from domestic prosperity. In a landscape where economic security is dictated by external capital inflows, systemic youth exclusion, and migration horizons, the domestic economic returns on high-level critical thinking and analytical literacy are profoundly depressed.

The Remittance Cushion and Economic Decoupling

Kosovo operates on a heavily remittance-reliant macroeconomic model, where personal financial inflows from the diaspora account for a staggering **17.3%** of total GDP. This structural reality creates a distortive feedback loop into educational incentives. For many families, baseline income and financial security are completely decoupled from local economic productivity or the academic performance of their children. When prosperity is secured via foreign bank transfers rather than competitive entry into a meritocratic domestic knowledge economy, the structural pressure on students to master complex cognitive tasks is deeply eroded, reducing schooling to a cultural ritual rather than an economic escalator.

The Emigration Horizon and the Export Credential

The arrival of official European Union visa liberalization in **2024** fundamentally altered the horizons of Kosovo's school-aged population, shifting aspirations from domestic career-building toward immediate migration. This has triggered an unprecedented brain drain, with skilled emigration rates historically exceeding **40%**. This structural exit horizon alters how schooling is utilized by households. Schools are not treated as incubation chambers to acquire complex cognitive tools for local advancement; instead, the system functions as a machine to obtain a credential (a baseline diploma) that acts as a ticket for legal export abroad. This explains the data paradox where students exhibit **85%** reading enthusiasm and an **87%** sense of school belonging alongside dismal scores. They value the diploma as social capital but face zero immediate incentive to convert that enthusiasm into actual cognitive mastery.

Structural Youth Exclusion and the NEET Crisis

For youth relying on the domestic market, the landscape is defined by systemic exclusion and low-cognitive-demand roles. The youth unemployment rate historically hovers between an alarming **30%** and **50%**, while the youth NEET rate sits at a critical **32%**. The domestic private sector is dominated by low-skill services, retail, and construction - industries that require routine compliance rather than advanced data synthesis or experimental reasoning. Simultaneously, coveted high-paying roles within the public sector have historically been influenced by political patronage rather than objective academic excellence. Consequently, students absorb a rational lesson: elite academic performance in primary reading, math, and science does not protect them from economic exclusion at home.

Budget Inflexibility and the Failure of Local Retainment

This labor market failure is reinforced by the state's inability to deploy its budget as a tool for economic transition. Because a rigid **90.6%** of the recurrent education budget is locked directly into wage lines, MESTI lacks the capital to build high-quality vocational tracking, tech-incubator partnerships, or modern science pipelines. The state cannot afford to fund localized learning-recovery or upskilling programs to shift the market out of this low-skill equilibrium. Deprived of public investment to tie advanced schooling to high-value domestic tech or engineering industries, the country remains trapped in a vicious cycle: low domestic returns discourage instructional transformation, ensuring that the graduating workforce lacks the cognitive capacity to attract advanced foreign direct investment.

Sources: World Bank Kosovo Human Capital Index Economic Brief (humancapital.worldbank.org) | IOM & UNDP Kosovo Migration and Brain Drain Survey | MESTI Macro-Labor Market Analysis Alignment Report.

Building Translational Infrastructure for Deeper Learning

As a young democracy that stabilized its civil infrastructure from zero post-1999, Kosovo's achievement of near-universal school enrollment and **87%** student belonging is a monumental success of national resilience. The transition from securing educational access to cultivating deep, 21st-century cognitive competency is an evolutionary process that requires time and structural patience. An effective path forward moves away from prescriptive top-down directives, focusing instead on open-ended policy directions that respect historical roots, enhance systemic oversight, and promote analytical learning.

Honoring the Historical Psyche of Resistance

Pedagogical modernization in Kosovo must respect the historical origins of its rote learning tradition. The contemporary reliance on memorization and dictation is deeply linked to the **1990s** parallel system, an era when citizens were banned from public school buildings and built underground networks in private basements. Operating entirely without textbooks, these makeshift spaces relied on one-way dictation to preserve literacy and national identity under siege. In that state of structural survival, memorizing the lesson became an act of patriotic resilience, while intellectual deviation was unconsciously viewed as a threat to collective solidarity. Modern policies should treat the transition to critical thinking not as correcting a flaw, but as the next honorable step in a history of educational resilience.

Shifting from Paper Compliance toward Systemic Accountability

To help teachers move past factual repetition, future policy frameworks could build supportive administrative bridges between legal texts and classroom realities. Kosovo's rapid legal adoption of an EU-style Competency-Based Curriculum demonstrates immense national ambition, but it has outpaced local school capabilities. Case analogs like Poland’s post-socialist overhauls suggest that successful systems prioritize non-punitive, localized mentoring over rigid compliance paperwork. By utilizing municipal education directorates to provide formative coaching and modeling higher-order questioning, the state can gradually equip teachers to guide students from basic retrieval toward independent evaluation.

Aligning Macro-Labor Demand with Analytical Skills

Shifting societal demand away from symbolic credentials requires an economic environment that visibly rewards critical reasoning. High-performing systems like Singapore demonstrate that student effort aligns closely with market incentives. When an economy actively compensates analytical skills, families naturally demand higher instructional rigor. While Kosovo’s **17.3%** remittance dependence can distort these incentives, future economic steering could look toward policies that reinforce high-value foreign direct investment and local technology employment. Creating domestic labor spaces requiring data synthesis and scientific inquiry provides a clear motivation for households to prioritize analytical mastery over a simple baseline diploma.

Operationalizing the Pre-Primary Foundation

Finally, protecting the long-term intent of recent compulsory early childhood mandates represents a vital lever to close the country's **80** to **85** point socioeconomic achievement gap. The legislative decision in the Education Strategy 2022–2026 to make pre-primary education mandatory was an exceptional, evidence-based choice. To prevent this mandate from colliding with existing infrastructure bottlenecks, capital allocations from international donors could prioritize retrofitting primary facilities into dedicated, play-based childhood environments. Insulating five-year-olds from desk-bound choral repetition protects early working memory development, allowing low-SES cohorts to enter Grade 1 ready for higher-order thinking.

Sources: IEA PIRLS 2021 & TIMSS 2023 Comprehensive National Reports | Kosovo Ministry of Education Strategy 2022–2026 (MESTI) | World Bank Group Transitional Education Policy Reviews (Poland/Singapore Case Analogs).